MIL-C-20610
15 January 1952

MILITARY SPECIFICATION

CLOTH, CAMBRIC, VARNISHED,
UNPARAFFINED

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army PXS-1185
24 January 1949

This specification consists of this cover sheet and Specification PXS-1185, dated 24 January 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
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PXS 1185
24 January 1949

PICATINNY ARSENAL TENTATIVE SPECIFICATION

CLOTH, CAMBRIC, VARNISHED, UNPARAFFINED

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of unparaffined varnished cambric cloth for use with ammunition.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications. - The following specifications of the issue in effect on the date of invitation for bids and all drawings and specifications listed or referred to therein, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN-P-223 Powder, Black

U.S. ARMY SPECIFICATION

100-2 Standard Specification for Marking Shipments by Contractors

(Army. -- Copies of Specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings. -- The following drawing of the issue in effect on the date of invitation for bids and all drawings and specifications listed or referred to thereon, form a part of this specification.

Picatinny Arsenal Drawing No. DP-44282 - 120° C Vacuum Stability

(Army. -- Copies of drawing should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Material. --
3.1.1 The fabric shall be a closely and uniformly woven cotton material, free from unsightly defects, dirt, knots, lumps and irregularities of twist.
3.1.2 The cloth, after varnishing, shall have a minimum count of 57 warp by 53 filling threads per inch.
3.1.3 The warp and filling threads shall run at right angles to each other in straight lines.

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3.1.4 A high-grade of slow-drying, clear baking varnish (oil base) shall be used.

3.2 The varnished cambric cloth shall consist of cotton fabric base thoroughly impregnated with linseed oil varnish.

3.3 Varnish.--The varnish shall be thoroughly baked and oxidized. Each coat of varnish shall be slowly dried at a low baking temperature.

3.4 Flexibility.--The varnish coating shall adhere well to the fabric, and shall not crack when the material is doubled upon itself (zero radius).

3.5 Color.--Unless otherwise specified, the color of the finished material shall be natural varnish color.

3.6 Finish.--The surface shall be smooth, clean, and free from porosity and visible defects.

3.7 Thickness.--Thickness shall be .010 $\pm$ .0015 inch.

3.8 Width.--The width shall be specified in the contract or order.

3.9 Length.--Unless otherwise specified the cloth shall be furnished in rolls containing 100 yards each.

3.10 Reactivity.--The reactivity shall be not more than 1.20 ml of gas. (See par. 4.3.2)

3.11 Moisture vapor transmission rate.--The moisture vapor transmission rate shall be .4 gram, maximum, per 100 square inches per 24 hours. (See par. 4.3.3)

3.12 Workmanship.--Workmanship shall be in accordance with high-grade commercial practice covering this class of material.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling.--One yard of the finished cloth shall be selected at random from each lot of 1000 yards of cloth or fraction thereof, on the contract.

4.2 Inspection.--

4.2.1 The place of inspection shall be as specified in the contract.

4.2.2 The material shall be subjected to a careful visual inspection to determine conformance with the requirements of this specification as to material, workmanship, and finish.

4.3 Tests.--

4.3.1 Determination of thickness.--The thickness shall be determined by the use of a micrometer having faces not less than 0.25-inch or more than 0.50 inch in diameter, or by use of an accurate dial gage measuring device acceptable to the inspector. At least ten readings shall be taken of thickness, and the maximum and average values recorded.

4.3.2 Reactivity test.--

4.3.2.1 Test procedure.--Select three heating tubes of the apparatus shown on Figure 1 and add to the first tube 2.5 grams of the cloth cut into small pieces about 3/32 inch by 3/32 inch square; add to the second tube 2.5 grams of black powder; and add to the third tube 2.5 grams of the cloth cut into pieces as above and 2.5 grams of black powder. To each tube, connect a capillary tube and seal the connection by means of 1 ml of mercury. Clamp the apparatus so that the long section of the capillary tube is in a nearly vertical position and the lower end seats on a solid support. Transfer 7.0 ml of mercury to the cup at the lower end of each capillary tube. By means of a vacuum pump, evacuate the system until pressure is reduced to approximately 5

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mm of mercury. Disconnect the pump and measure the total vertical height of the columns of mercury in each capillary tube. Measure and subtract the respective vertical heights of the mercury in the cups. The difference is the value, H_1. Note the temperature (t_1) and the barometric pressure. Insert the tubes containing black powder and the cloth only, in the constant temperature bath, shown on Drawing DP-44282, containing a mixture of glycerine and water having a specific gravity of approximately 1.05 which will maintain a temperature of 100.0° ± 0.5° C for 40 hours. Remove the heating tubes and allow to cool to room temperature. Measure the vertical height of the mercury column in each capillary tube and subtract from each the respective vertical height of the mercury in the cups. The difference is the value H. Note the room temperature (t) and the barometric pressure. Calculate the volume of gas at standard temperature and pressure, liberated during each test, by means of the following formula:

$$\text{Volume of gas in ml} = A + C(B - H)\frac{273P}{760(273 + t)} - A + C(B_1 - H_1)\frac{273P1}{760(273 + t_1)}$$

where:

A= volume of the heating tube minus v (volume of the particular sample of explosives and/or compound added to the tube).
B_1= total length of capillary tube (in mm) minus the vertical height of mercury in the cups before heating.
B= total length of capillary tube (in mm) minus the vertical height of mercury in the cup at the end of the test.
C= the determined unit capacity of the capillary tube in ml per mm. (See Par. 4.3.2.3)
P_1= the barometer reading at the start of the test minus H_1.
P = the barometer reading at the end of the test minus H.

4.3.2.2 Calculation. -- Calculate the reactivity of the cloth in contact with the black powder as follows:

$$\text{Reactivity in ml gas} = X \quad (Y + Z)$$

Where:
X = ml gas produced by the mixture of black powder and cloth.
Y = ml gas produced by the black powder alone.
Z - ml gas produced by the cloth alone.

4.3.2.3 Calibration of glass tubes. -- Determine the volume in ml of the 15.5 cm. heating tube by running mercury from a burette into the tube until it is filled to the level at which the ground glass joint of the capillary tube will make contact with the mercury. Transfer 7.0 ml of mercury to the cup at the lower end of the capillary tube, clamp the tube in an upright vertical position and measure the height in mm of the mercury column in the capillary tube. Measure the length in mm of each of the 3 parts of the capillary tube, add these values and subtract the height of the mercury column (approximately 25 mm) determined as directed above. Determine the capacity of the capillary

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tube per unit of length as follows: Transfer an accurately weighed sample of approximately 10 grams of mercury to the cup at the lower end of the capillary tube. Manipulate the tube so that, when the tube is horizontal, mercury is contained in a continuous section of the longest part of the tube. Measure the length of this mercury column. Repeat this procedure twice with the mercury in 2 other parts of the largest section of the tube. Average the 3 lengths measured and calculate the unit capacity of the capillary tube as follows:

$$\text{ml per mm} = \frac{W}{13.59L}$$

Where:

W = weight of mercury used.
L = average length of the mercury column in mm.

4.3.3 Moisture-vapor transmission rate --

4.3.3.1 Preparation of specimen.--A smooth 0.0508-inch (16 gage) aluminum disc 5 7/8 inches in diameter shall be provided with an opening 1 1/2 by 4 inches cut in the center. A piece of cloth 4 1/2 inches long and 2 inches in width shall be applied over the 1 1/2 by 4-inch opening of the disc, so that it is in contact with 1/4 inch of each of the edges surrounding the opening. (For cloth samples less than 2 inches wide, the width of opening in the disc shall be reduced accordingly.) The cloth shall be wax-sealed to the disc using the wax mixture prepared as described in paragraph 4.3.3.3.2. The wax shall be applied to the cloth in such a manner as to completely seal the edges of the cloth to the aluminum disc and cover all the cloth except the part directly over the opening with a layer of wax. The aluminum disc, with the cloth side up shall be wax-sealed over the calcium chloride in the Southwick test dish and exposed in the humidity cabinet.

4.3.3.2 Apparatus.--

4.3.3.2.1 Humidity cabinet.--The humidity cabinet to be used in this test is standard equipment known as the General Foods moisture-vapor transmission cabinet. It shall provide a relative humidity of 90 $\pm$ 2 percent at a temperature of 37.1° $\pm$ 0.3°C (98.8° $\pm$ 0.5°F) with no condensation on the test dishes or in the space in which the test dishes are placed. The circulation over the test dishes shall be negligible.

4.3.3.2.2 Test dishes and brass template.--The test dishes and brass template shall conform to the standard size to be used in the standard General Foods moisture-vapor transmission cabinet.

4.3.3.3. Procedure.--

4.3.3.3.1 A 50-ml beaker shall be filled with 8-mesh anhydrous calcium chloride. The calcium chloride shall be spread evenly over the bottom surface of the dish and the test specimen placed over the calcium chloride concentric with the rim of the dish. The brass template shall be carefully placed over the aluminum disc to which the cloth being tested has been applied.

4.3.3.3.2 A wax mixture consisting of 60-percent amorphous wax and 40-percent paraffin wax shall be heated in a porcelain dish to at least 100°C (212°F). It shall then be poured through a 20-mesh screen to remove any large

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pieces of foreign matter that may be in the wax. The wax mixture shall be poured into the annular space between the template and the rim of the test dish filling up this space to approximately flush with the top of the template. The dish shall then be cooled to harden the wax sufficiently so that the template can be removed. Care should be taken not to cool the dish so long as to make the wax hard and brittle. The template shall be removed by inserting a screw driver successively under each ear of the template and giving a slight twist, tending to press the wax against the dish and at the same time raising the template. Any difficulty experienced in extracting the template from the dish can be overcome by first rubbing the edge of the template with vaseline before pouring wax.

4.3.3.3.3 After the template has been removed, the dish shall be inspected for loose pieces of wax, etc., and examined for flaws in the seal. It shall then be placed in the humidity cabinet for conditioning for a period of 24 hours. It shall then be removed from the humidity cabinet, cooled for 5 minutes in a room maintained at 73° ± 3.5°F with 50 percent relative humidity, weighed on an analytical balance, and returned to the humidity cabinet. The procedure prescribed in this paragraph shall be repeated, with exposure periods of not less than 48 hours each, until the transmission rate becomes constant.

4.3.3.3.4 After the moisture-transmission rate becomes constant, it shall be reported in grams of moisture per 100 square inches of tape area for 24 hours, and shall be calculated as follows:

$$\text{Transmission rate} = \frac{(W_2 - W_1) \times 2400}{T \times A}$$

Where:

W_1 = weight (in grams) before exposure period.
W_2 = weight (in grams) after exposure period.
T = exposure period (in hours).
A = area (in square inches) of opening disc.

minimum of three tests shall be made and the average of all tests on a sample shall be reported.

5. PREPARATION FOR DELIVERY

5.1 Packaging.--The rolls of cloth shall be packed in individual cartons. Both the roll and the carton shall be tagged or labeled with the following information:

Cloth, Cambric, Varnished, Unparaffined.
Specification: PXS-1185
Width_____ Thickness_____
Quantity_____ (Yards)
Order No._____
(Mfr's. Name or Trademark)

5.2 Packing.--Unless otherwise specified the cartons shall be packed in

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standard commercial containers so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

5.3 Labeling and marking.--

5.3.1 Marking for shipment shall be in accordance with U. S. Army Standard Specification for Marking Shipments by Contractors No. 100-2.

5.3.2 Labels.--If labels are used, they shall be placed on cartons in such a manner as not to interfere with breaking the seals on the cartons.

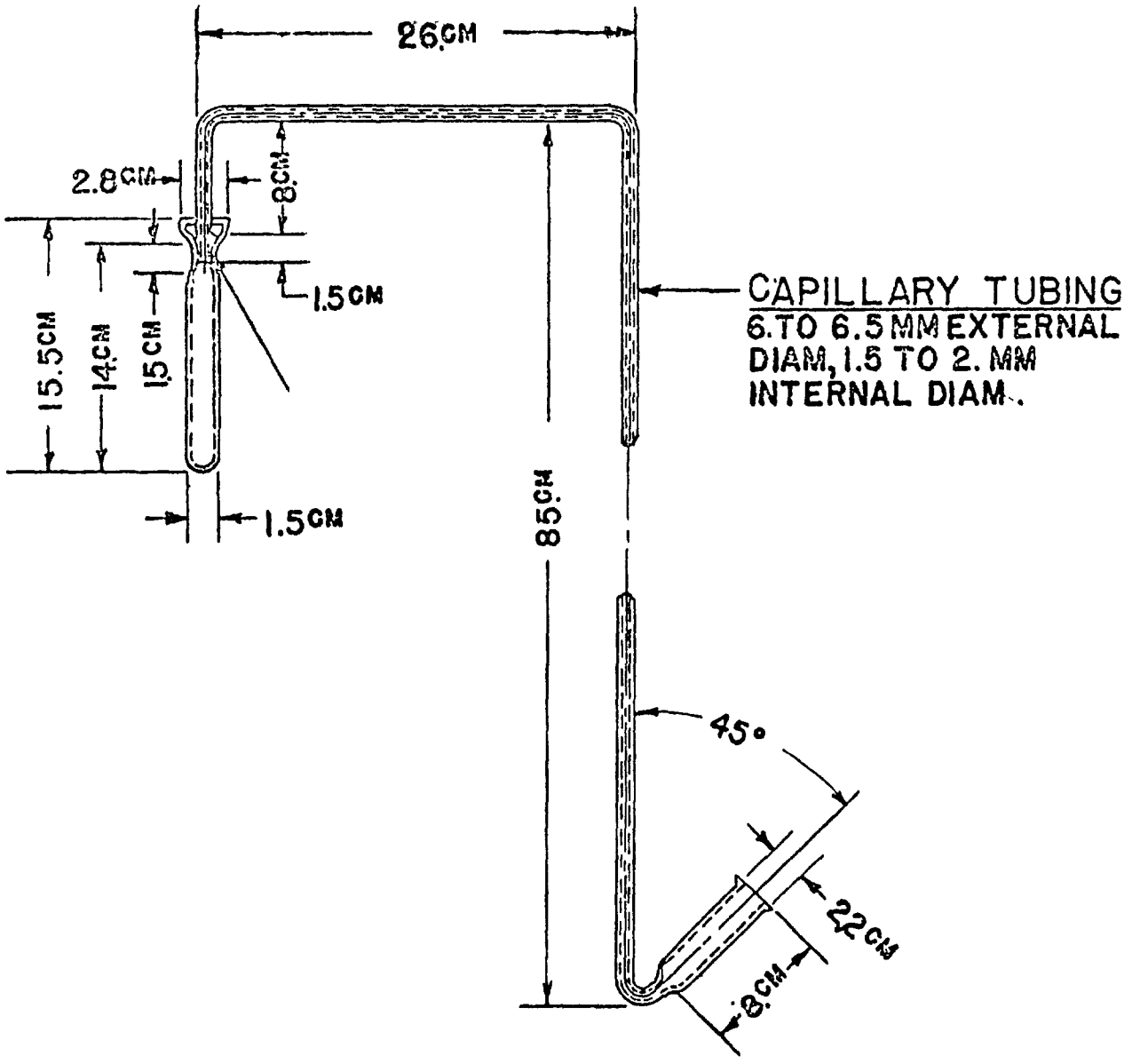
6. NOTES

6.1 Information concerning tests to demonstrate acceptability of unparaffined varnished cambric cloth for use in ammunition items may be obtained from the Office, Chief of Ordnance, Washington 25, D. C. or from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey.

6.2 NOTICE.--When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.3 Note.--Copies of this specification may be obtained from Commanding Officer, Picatinny Arsenal, Dover, New Jersey.

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APPARATUS FOR
100°C VACUUM STABILITY TEST

FIG 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

PECIFICATION

RGANIZATION *(Of submitter)* | CITY AND STATE

ONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

ATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

BMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1426
APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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